

Beyond All-Equity Lifecycle Portfolios: A Public-Data Multi-Asset Extension

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Abstract

Should lifecycle investors hold only equities? Anarkulova, Cederburg, and O'Doherty (2025) find that internationally diversified stocks outperform conventional balanced and target-date allocations in their baseline model. We recover the domestic-to-international equity ranking in an annual public-data implementation, then widen the investment universe to currency-hedged global bonds, gold, and trend-following futures. Two simple four-sleeve families use fixed weights at exposures from 100% to 200%. Over 1927–2025, the equal-weight rule at 175% exposure reduces simulated retirement ruin from 6.98% to 2.79% and matches benchmark utility on 7.16% of income instead of 10%. Under each of the three specified source influence diagnostics, this rule has lower annual volatility, lower ruin and a lower saving requirement than the benchmark. Both diversified families already reduce ruin without portfolio-level borrowing, although matching benchmark utility then requires more saving. Later historical windows also support the diversified ranking. The evidence supports broadening the opportunity set when evaluating fixed lifecycle allocations. It remains conditional on the return proxies, annual timing, and financing and currency-hedging assumptions.

JEL classifications: C15, D14, G11, G17.

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1 Introduction

Households saving for retirement face two distinct decisions: which assets to combine and how much exposure to take. Balanced and target-date portfolios often change both at once. Adding bonds reduces the equity share and usually reduces total volatility, so the effects of composition and exposure arrive together. Lifecycle models place these choices in a household setting with labor income, longevity risk, retirement consumption, and bequests (e.g., Campbell and Viceira, 2002; Cocco, Gomes, and Maenhout, 2005). Anarkulova, Cederburg, and O’Doherty (2025), hereafter ACO, find that a fixed allocation of approximately 33% domestic and 67% international equities outperforms conventional balanced and target-date strategies in their baseline setting. Their opportunity set contains domestic and international stocks, local sovereign bonds, and local bills. Their leverage sensitivities also show that financing costs influence the optimal allocation.

How far does the equity result extend? ACO diversify stocks internationally but their baseline bond allocation holds local sovereign debt. We broaden the bond sleeve across issuers and add gold and trend-following futures. These assets bring different exposures to inflation, currency movements and market trends. Whether those differences improve retirement outcomes depends on their joint return distribution and on the cost of holding them.

The extension follows the distinction between portfolio composition and total exposure in Markowitz (1952) and Asness (1996). Diversification can improve the distribution of returns available to a household, while borrowing scales both gains and losses and incurs financing costs. We evaluate these choices together through two fixed portfolio families. Their annual volatility is a reported outcome of the comparison, alongside ruin and expected utility.

Our annual replication uses the Jordà–Schularick–Taylor Macrohistory Database and supplementary public sources.¹ The final panel contains 1,561 country-years across 16 developed markets over 1927–2025. It recovers the qualitative ranking of the fixed equity and balanced strategies: simulated retirement ruin is 6.98% for the 33/67 benchmark, compared with 15.52% for domestic equities. Differences in data, frequency, and simulation conventions make this a comparison within a public data implementation, rather than an exact reproduction of ACO’s estimates.

The central experiment keeps the benchmark’s equity mix fixed and adds the three other sleeves in two families: one proportional to 60/40/25/25 and one equally weighted. Each family is evaluated at gross exposures of 100, 125, 150, 175, and 200%. The portfolio weights are fixed heuristics, not estimates of historical means, covariances, or optimal utility. This avoids fitting weights to sample moments, although the choice of sleeves and strategy families remains a research decision and was not independently pre-registered.

The full-panel equal-weight allocation at 175% exposure reduces simulated ruin from 6.98% to 2.79% and equivalent saving from 10% to 7.16%. Under all three specified source-influence diagnostics, it has lower annual volatility as well as lower ruin and equivalent saving. Removing Italy 1942 gives 17.30% volatility against 18.17%, 2.61% ruin against 7.19%, and 7.02% equivalent

¹The research repository is available at <https://github.com/GF-boop/beyond-stocks>.

saving against 10%. Removing France 1946 as well, or Italy throughout the sample, gives the same direction on all three measures. These diagnostics were chosen after examining the influential observations; they establish sensitivity to those exclusions, not independent validation.

The primary sample remains 1927–2025. Extreme returns alone are not grounds for deletion, and excluding early history would remove adverse regimes as well as questionable conversions. Post-1950 and post-1970 windows provide explicit sensitivities, selected during the methodological audit rather than independent validation samples. Their results preserve the favorable outcome ranking of the two 200% families. The analysis remains conditional on the synthetic managed-futures proxy, historical gold regimes, annual timing and assumed financing and currency-hedging conventions.

2 ACO and the Multi-Asset Extension

2.1 The benchmark result

ACO evaluate retirement strategies in a lifecycle model with stochastic labor income, Social Security benefits, mortality risk, and a bequest motive. A couple works for 40 years and then draws down its retirement account. Their monthly return panel covers 39 developed countries over 1890–2023, with coverage varying by country. Multi-year bootstrap blocks can originate in different countries, while household earnings, demographics, and preferences remain anchored to a U.S. baseline.

Their baseline fixed allocation holds approximately 33% domestic and 67% international equities without borrowing. ACO also relax the leverage constraint: the optimal exposure and bond allocation depend on the borrowing spread. Our extension examines additional asset classes under explicit financing assumptions, whose sensitivity is reported in Section 5.

ACO select portfolio weights by maximizing average simulated utility. On our public panel, a grid search restricted to the domestic–international equity mix places the optimum at 25% domestic and 75% international, with 33/67 within 0.02 percentage points of equivalent saving. We therefore retain 33/67 as a common equity sleeve and reference strategy. The multi-asset families use fixed heuristic weights rather than an analogous optimization. This comparison evaluates specified alternatives to the equity benchmark; it is not an out-of-sample test of either allocation.

Expected utility already trades off favorable outcomes against adverse ones under the household’s preferences. Comparing portfolios at similar annual volatility adds a complementary diagnostic: it helps distinguish changes in composition from changes in overall exposure. We use the fixed ladders of Section 5 for that purpose. Their realized volatilities are outcomes of the experiment, not calibration targets, and similar annual volatility does not imply identical tail or lifecycle risk. Appendix E separately examines how the comparison changes with risk aversion.

2.2 Three empirical questions

Our investigation addresses three questions.

First, how much of the international-equity comparison survives when real exchange rates are held constant? We keep the national markets and weights fixed and remove the joint movement of

nominal currencies and relative prices. The counterfactual separates the accounting contribution of currency movements; it does not represent an investable hedge or identify a causal effect.

Second, does replacing local sovereign bonds with a currency-hedged global bond basket improve outcomes? Keeping equity weights and leverage fixed allows us to compare the two bond sleeves under the stated currency and financing conventions.

Third, do the four-sleeve portfolios improve retirement outcomes across fixed exposure levels? We compare ruin, equivalent saving and annual volatility, then examine which conclusions survive changes in the historical sample, household preferences and implementation costs. This last step matters because extreme currency conversions can move the volatility comparison without reversing the lifecycle ranking.

3 Data

3.1 Country-year panel and resident numeraire

All empirical inputs to our lifecycle simulations are publicly accessible. Whereas ACO evaluate their model on proprietary monthly return series, we reconstruct the long-run developed-market record from open sources, ensuring that the replication and every extension can be reproduced from public data alone. The only proprietary series in the paper are the SG and BarclayHedge managed-futures indexes, which appear solely in the external validation of Appendix B.2 and enter no lifecycle simulation.

The panel begins with release 6 of the Jordà–Schularick–Taylor Macrohistory Database (Jordà et al., 2023; Jordà et al., 2019), which supplies nominal total returns on equities and long-term government bonds, short rates, inflation, exchange rates, and real GDP for 16 developed markets through 2020. We extend each series through 2025 using dividend-adjusted local indices, locally listed ETFs, and the Global Macro Database (Müller et al., 2025). Over the 2006–2020 overlap, the reconstructed equity series average approximately 1.2 percentage points below the Macrohistory series, primarily reflecting ETF management fees and large-cap benchmark tilts. A source indicator flags the appended years, allowing robustness checks to isolate the original Macrohistory window. The resulting baseline dataset is an unbalanced panel of 1,561 country-years.

An ex ante investability screen is retained as a labeled sensitivity. It excludes calendar years overlapping the prolonged closures or restrictions reported for our countries in ACO’s Table A.III and years with CPI inflation of at least 50% or deflation of at least 20% in absolute value. These criteria identify observations for which an annual vector cannot synchronize a tradable market, a convertible currency, and a renewable hedge. None of them depends on portfolio returns. The replication package preserves the raw rows. Every calendar year from 1927 through 2025 remains represented.

Let $S_{c,f,t}$ denote units of resident currency c per unit of foreign currency f . If $R_{f,t}^N$ is a nominal

foreign return and $\pi_{c,t}$ is resident inflation, the resident-real return is

$$1 + R_{f,t}^{(c)} = (1 + R_{f,t}^N) \frac{S_{c,f,t}}{S_{c,f,t-1}} \frac{1}{1 + \pi_{c,t}}. \quad (1)$$

We apply this spot conversion to unhedged assets before aggregating returns. Domestic equities, international equities, gold, and unhedged controls therefore all measure changes in purchasing power within the resident household’s currency.

International equity excludes the country of residence and uses lagged market capitalization weights constructed from the Big Bang Database’s historical market-capitalization-to-GDP ratios when coverage is sufficient (Kuvshinov and Zimmermann, 2022), with real-GDP weights (Maddison Project via the Macroeconomic History Database, extended by Müller et al. 2025) as a fallback. The 33/67 benchmark combines the resident’s domestic market with this foreign basket. The constant-real-exchange-rate counterfactual instead aggregates each foreign market’s local real return with identical weights. It is an accounting exercise on the same country-year states. No investor could have held it: currency carry and hedging costs fall outside its scope.

The world bond block equally weights every sovereign in the panel with a usable bond, bill, and exchange rate in a given year. In 1,344 of 1,561 country-years it holds all 16 issuers; the floor anywhere is 13. Years with fewer than eight available issuers are dropped, which costs no observation after 1927. Like an actual global bond index, the block includes the resident’s own sovereign. The basket is therefore common to all residents in a year, and each foreign bond uses the initial-notional forward of equation (4). Financing stays local throughout. We subtract ten basis points annually from covered notionals in the baseline. The unhedged foreign bond and bill basket is retained only as a labeled diagnostic control. Wider than a four-issuer basket, the resulting block still covers developed markets alone, so it is not exhaustive global fixed income.

3.2 Gold and managed futures

Gold uses the annual U.S.-dollar price compiled by MeasuringWorth (Officer and Williamson, 2024) through 1967 and the LBMA afternoon fixing (London Bullion Market Association, 2025) from 1968, the first year a free market existed. Both are dollars per troy ounce, spliced without recalibration. We subtract 40 basis points per year for custody, convert the dollar return into each resident currency, and deflate by resident inflation. One caveat governs everything the paper says about gold. For much of the first 41 years the dollar price was administered, and private U.S. ownership and convertibility were restricted. This is not a U.S. peculiarity: under the gold standard and the Bretton Woods pegs every panel currency’s gold price was set through its parity, so the low-variance segment reflects the monetary regime. The price series does not establish what a household could have earned under ownership and conversion restrictions. It is therefore an uncertain basis for estimating prospective gold risk: the variance was suppressed by policy, and we would not want a reader to take the pre-1968 segment as evidence about gold in a free market. We therefore read the gold results as sensitivity evidence, not structural estimates (Erb and Harvey,

2013).

No free investable managed-futures index spans the sample. We reconstruct a monthly time-series-momentum proxy instead, following a well-established literature. Moskowitz, Ooi, and Pedersen (2012) document that the sign of an asset’s own past twelve-month return predicts its next-month return across roughly 58 futures markets, and that scaling each position by inverse volatility and averaging produces a diversified premium. Hurst, Ooi, and Pedersen (2017) extend the same rule back to 1880 and show that a blend of one-, three-, and twelve-month look-backs is stable across a century of data and across asset classes. Nick Baltas and Kosowski (2020) turn to the construction choices themselves. Signal horizon, rebalancing frequency, and the resulting turnover jointly determine net performance and capacity. Hence our one-, six-, and twelve-month blend, and hence the explicit turnover cost below. Lempérière et al. (2014) find a trend premium in equities, bonds, currencies, and commodities as far back as 1800. That history is what justifies running the rule over the full 1927–2025 window. Trend following also lowers sequence risk in retirement decumulation (Clare et al., 2017; Clare et al., 2019) and improves risk-parity implementations through correlation-aware weighting (Nikitas Baltas, 2015). That literature, however, stops at portfolio-level backtests: none of it embeds trend following in a full lifecycle model with the pooled developed-market panel, which is the gap the lifecycle results of Section 5 fill.

The word “monthly” refers solely to the construction of this sleeve. It does *not* mean that we interpolate the annual JST/Macrohistory panel used for domestic equity, bonds, bills, inflation, and the bootstrap. The MF signal instead reads a separate, frozen monthly price snapshot, and only its completed calendar-year returns are then joined to the annual household panel. Appendix B.1 documents that snapshot, its sources, and the monthly-to-annual transformation in detail.

In our implementation, signals average the signs of prior one-, six-, and twelve-month returns. Volatility comes from the preceding 36 months. The aggregate strategy targets 10% volatility under a three-times scalar cap. The universe expands across four sectors as data become available: 18 national price indexes and 18 synthetic ten-year sovereign positions, 19 commodities plus gold and silver, and six dollar foreign-exchange forwards. The monthly equity and yield inputs come principally from the OECD Main Economic Indicators via FRED (Organisation for Economic Co-operation and Development, 2026; Federal Reserve Bank of St. Louis, 2026). The early U.S., U.K., German, French, commodity, and rate segments come from the monthly NBER Macrohistory files (National Bureau of Economic Research, 2016), commodities are extended with the World Bank “Pink Sheet” (World Bank, 2025), and currencies use month-end Federal Reserve H.10 quotations (Board of Governors of the Federal Reserve System, 2026). Where two sources meet, the return that straddles the junction is left missing. Computing it from spliced price levels would fabricate a price change that never happened. Sector Sharpe ratios over the full reconstruction are 0.44 (equities), 0.50 (rates), 0.47 (currencies), and 0.36 (commodities), against 0.41, 0.49, 0.57, and 0.80 in the Capital Fund Management estimates of Lempérière et al. (2014). Combined, the figures are 0.77 against 0.80.

The proxy includes the U.S.-dollar collateral return recorded in the snapshot. The baseline treats

Table 1: Realized resident-real portfolio properties

Portfolio	Mean return	Volatility	1% annual return
Domestic equity	7.60%	23.34%	-45.64%
ACO 33/67	8.12%	25.28%	-40.36%
Stocks/I 50/50	7.99%	22.59%	-40.22%
Balanced domestic	5.38%	15.82%	-34.19%
Balanced/I	5.61%	15.04%	-27.63%
90/60 local	8.25%	23.95%	-36.55%
90/60 global bonds	8.19%	24.13%	-36.19%
ACO, constant real FX	7.44%	15.75%	-37.40%
60/40 ACO/covered	5.64%	16.51%	-25.67%
60/40 ACO + 33.33 MF	7.60%	17.90%	-21.66%
60/40 ACO + 33.33 gold	6.79%	23.86%	-23.33%
54/36/20/20 ACO	7.00%	19.87%	-19.67%
90/60/25/25 ACO	10.52%	30.27%	-29.94%

Note: Arithmetic means, standard deviations and linearly interpolated 1% quantiles of pooled annual returns across 1,561 country-years. Source: `results/main_core_n10000.json`. These are annual-return moments, not drawdowns or confidence intervals. Bonds and managed futures use the fixed-notional hedge; managed futures include baseline fees and costs.

its dollar NAV as a foreign asset and applies the initial-notional forward in equation (4), using the recorded collateral return for the foreign cash leg. Currency exposure on subsequent NAV gains or losses remains. This is the same hedge convention as for global bonds. Replacing collateral for the entire terminal value is the alternative full-value convention. A locally collateralized futures strategy would be a different implementation. We subtract a 0.85% annual management fee and transaction costs based on measured turnover of 18.88 times per year and a three-basis-point spread, or 0.57% annually. Its monthly correlation with several official CTA indexes is 0.49–0.53 over 2000–2025.

3.3 Realized portfolio properties

Table 1 reports pooled one-year return moments from the same corrected panel and hedge convention as the lifecycle results. These statistics describe the evaluated rules; they do not establish equal tail or lifecycle risk. Global bonds have slightly higher volatility than local bonds in the 90/60 comparison under this convention.

4 Lifecycle Model and Portfolio Design

4.1 Household, income, and retirement

The household is a couple aged 25. Each spouse receives an independent stochastic earnings history through age 64 based on model 6 of Guvenen et al. (2021). Log income contains a quadratic age profile, a persistent component with autoregressive coefficient 0.959, transitory shocks, and age- and income-dependent nonemployment. The household saves a constant fraction of each spouse’s

income when that individual’s annual income is at least \$15,000. Contributions arrive at the start of the year, in time to earn that year’s portfolio return.

At age 65 the couple fixes a real annual withdrawal equal to 4% of retirement wealth, following Bengen (1994) and ACO. A fixed real withdrawal makes the retirement outcome sensitive to the order in which returns arrive, not only to their average: a sequence of early losses can exhaust the account even when long-run returns are adequate, and international evidence shows the safe rate varies widely across historical return paths (Estrada, 2018). Appendix F varies the withdrawal rate from 3% to 5% and the contribution rate from 5% to 15% and finds the diversified ranking unchanged. Social Security follows from each simulated earnings history, computed with the 2022 \$147,000 taxable-earnings ceiling, 2022 bend points, spouse and survivor benefits, and the early-claiming adjustments ACO describe. Supplemental Security Income supplies a consumption floor. Ruin occurs when financial wealth cannot fund the fixed withdrawal before the last spouse dies. Public pension income continues after financial wealth is exhausted.

Mortality follows sex-specific Gompertz–Makeham laws calibrated to the joint retirement and longevity moments reported by ACO from Social Security Administration tables (Social Security Administration, 2020). Return paths, income histories, and death draws are paired across strategies.

This household combines U.S. household assumptions with pooled financial history, following ACO’s general design. The combination deserves to be named plainly. Earnings, Social Security, Supplemental Security Income, and preferences are U.S. throughout. The return-and-inflation path is resampled from the pooled developed-market panel. From one block to the next it can move across countries, and so across numeraires. A simulated lifetime is not a resident of anywhere: it is a U.S. household drawing a sequence of macro-financial decades from sixteen developed markets. The premise is that one national record is a single historical path, too thin to describe a tail. The cost of that premise is real: a simulated lifetime can splice a high-inflation 1970s decade in one country onto a deflationary decade in another. Pooling adds historical environments, but countries share shocks and global asset baskets: the 1,561 country-years are not independent observations. Every strategy faces the same pooled environment within each experiment. Section 4.4 gives the mechanics.

4.2 Utility and equivalent saving

For annual real consumption C_t , household size H_t , bequest B , and last year alive T , utility is

$$U(C, B) = \sum_{t=65}^T \frac{(C_t/\sqrt{H_t})^{1-\gamma}}{1-\gamma} + \theta \frac{(B+k)^{1-\gamma}}{1-\gamma}. \quad (2)$$

Risk aversion is $\gamma = 3.84$, the bequest motive has intensity $\theta = 2,360$, and the shifter is $k = \$490,000$. We chose none of the three. They are the estimates of De Nardi, French, and Jones (2010), fitted to retiree data in a lifecycle model with medical expenses, and ACO adopt them unchanged. Our utility aggregator therefore uses their stated preference parameters, while the wealth and consumption dynamics remain an annual approximation of their monthly engine. The

Table 2: Fixed portfolio rules

Portfolio	Equity	Bonds	Gold	MF	Sleeve sum	Financing
ACO 33/67	100% ^a	0%	0%	0%	100%	0%
90/60 local	90 ^a	60 ^b	0	0	150	50
90/60 global bonds, covered	90 ^a	60 ^c	0	0	150	50
60/40 ACO/covered	60 ^a	40 ^c	0	0	100	0
60/40 ACO + 33.33 MF	60 ^a	40 ^c	0	33.33	133.33	33.33
60/40 ACO + 33.33 gold	60 ^a	40 ^c	33.33	0	133.33	33.33
54/36/20/20 ACO	54 ^a	36 ^c	20	20	130	30
90/60/25/25 ACO	90 ^a	60 ^c	25	25	200	100

Note: Percentages of investor capital. ^a33% domestic and 67% international equities. ^bResidence-country government bond and bill. ^cCurrency-hedged equal-weight global sovereign bonds with resident-currency financing. The equity sleeve remains 33/67. “Sleeve sum” is the sum of index-level allocations, and financing is the corresponding external cash financing in the return equation.

bequest term is additive and warm-glow: utility depends on the size of the estate left, $B + k$, not on the heirs’ own circumstances, which is the specification that explains why retirees in the data run down wealth slowly. ACO evaluate monthly consumption and therefore multiply the annual bequest-intensity estimate by 12^γ . If within-year consumption is uniform, dividing their entire monthly utility by 12^γ gives equation (2) with $\theta = 2,360$. Under uniform within-year consumption, the annual aggregation thus preserves the relative scale of consumption and bequest utility.

For strategy j , the equivalent saving rate s_j^* solves

$$E[U_j \mid s_j^*] = E[U_{33/67} \mid s = 0.10]. \quad (3)$$

A rate below 10% means the strategy reaches benchmark utility on less saving. Every other reported outcome, wealth, consumption, ruin, and bequest, uses the common 10% rate and not s_j^* . The metric judges the allocation as a whole, composition and total volatility together. That is also what it cannot do. A quieter portfolio can need a higher equivalent saving rate purely because it takes less market exposure. The fixed-exposure ladders in Section 5 and Appendix C therefore report utility alongside annual volatility. This comparison helps describe the role of exposure; it does not hold every dimension of risk constant.

4.3 Fixed, non-estimated portfolio rules

Table 2 defines the main extensions. The 90/60 local strategy holds 90% in the ACO 33/67 equity portfolio and 60% in the residence-country sovereign bond, financed with a 50% short local-bill position plus the financing spread ϕ on the borrowed portion. The intermediate strategy changes only the bond sleeve to the covered global aggregate, with financing still in the resident bill. Every other extension also uses the same 33/67 resident equity return. The designs therefore vary fixed income, alternative sleeves, and total exposure without changing equity geography.

The weights are design inputs fixed prior to evaluation. While this discipline prevents direct

full-sample leverage optimization, sleeve selection still constitutes a modeling choice.

The manifest also contains two fixed leverage ladders, each stepping a four-sleeve family through 100, 125, 150, 175, and 200% gross exposure: one proportional to 60/40/25/25 (equity/covered bonds/gold/managed futures), the other equal-weight across the four sleeves. The levels are fixed before evaluation and target no realized volatility. These ladders carry the central comparison in Section 5.

Let A_t^f be the foreign asset's gross nominal return, B_t^f and B_t^c the foreign and resident gross nominal bill returns, and $q_t = S_{c,f,t}/S_{c,f,t-1}$ the spot currency movement. The baseline covers the initial foreign principal with a one-period forward. Under the assumed covered-interest-parity forward ratio, its resident-real return is

$$1 + R_t^{c,H} = \frac{B_t^c/B_t^f + (A_t^f - 1)q_t}{1 + \pi_{c,t}}. \quad (4)$$

The asset gain or loss beyond the initial principal retains currency exposure. Annual realized bill returns proxy the interest rates used to price the forward; this is a modeling convention, not a historical record of executable forward quotes. The former full-value hedge is retained only as a sensitivity. Let every $R_{i,t}^{(c)}$ then be in the common resident-real numeraire. A portfolio whose index-level sleeve sum is G_j has return

$$R_{j,t}^{(c)} = \sum_i w_i R_{i,t}^{(c)} + (1 - G_j) R_{f,t}^{(c)} - \max(G_j - 1, 0) \phi - h_j \kappa, \quad (5)$$

where ϕ is the financing spread, h_j is covered notional, and κ is the hedge friction. The managed-futures NAV includes its recorded U.S. collateral once; the forward payoff is added under equation (4). The portfolio financing position in equation (5) is separate.

For a levered portfolio ($G_j > 1$), the coefficient on the resident bill in equation (5) is negative. Holding all asset returns fixed, a higher ex-post real bill therefore lowers the portfolio return, while an inflation surprise that lowers the ex-post real bill raises it. This effect follows directly from the accounting identity of annual rebalancing, abstracting from intra-year borrowing constraints.

The baseline financing and hedge charges are 30 and 10 basis points, respectively. They are assumed annual real-return deductions, not estimates of historical execution costs or quoted nominal borrowing spreads. The latter would require deflation by resident inflation. Cost sensitivities vary these assumed deductions while retaining the observed real bill return.

Gross exposure here sums allocations to return sleeves. It does not measure all underlying derivative notionals: managed futures itself uses variable exposures. Packaged funds are a possible implementation route, but the study does not establish an instrument-level replication. Internal deleveraging, intrayear collateral calls, tracking costs and changing borrowing terms may produce outcomes different from annually rebalanced synthetic portfolios. No claim about the absence of funding risk follows from the annual simulation.

4.4 Bootstrap and inference

The main fixed-portfolio experiment draws 10,000 paths using a stationary block bootstrap with mean block length ten years, following ACO. Ten-year blocks preserve two things an i.i.d. resampling would distort: the horizon-dependent structure of equity returns, including the long-run mean reversion documented by Poterba and Summers (1988), and the persistence of real interest rates. The block advances through consecutive years of a country’s history, preserving the joint asset-return vector. A completed block starts again at a uniformly sampled panel row. At a gap or series end, the unfinished block continues at the first available year of another country. The implementation selects that country through a uniformly sampled row, so countries are weighted by their available sample lengths. This annual implementation follows the continuation principle of ACO; it is not an unchanged monthly replication. Section 4.1 discusses its interpretation.

Our resident-numeraire conversion operates inside each block. Every asset on a country-year row already sits in that country’s currency and price index (Section 3). A block drawn from, say, Japan in the 1980s is therefore a fully consistent yen-real decade: domestic and foreign equity, the hedged bond and managed-futures blocks, and inflation all measured in one household’s purchasing power. The numeraire changes only when the next block comes from another country. This is one coherent way to define the pooled distribution, and not the only valid one. A fixed-numeraire construction instead converts every source state into that numeraire. Such a control requires rebuilding the hedge in the target currency and is not reported in the corrected results below. Both constructions remain synthetic, for the reason Section 4.1 gives.

We report normal 95% intervals for the paired difference in ruin relative to the 33/67 benchmark. These intervals measure Monte Carlo uncertainty conditional on the observed panel. They do not treat 1,561 country-years as independent observations. Appendix D separately perturbs the historical sample itself, and its ranges are sensitivity diagnostics. We do not present them as confidence intervals.

5 Results

5.1 Replication and the central comparison

The corrected annual panel contains 1,561 country-years. We use the stationary bootstrap with the continuation rule described by Anarkulova, Cederburg, and O’Doherty (2025), retain historical observations except for documented market-closure and monetary-reform exclusions, and value the currency hedge as a one-period, fixed-notional forward. These choices are material: a return-dependent screen used in an earlier draft is not part of the reported baseline.

Table 3 gives the primary results. The public-data replication preserves the economic direction of the international-equity comparison: ACO 33/67 has lower simulated ruin than domestic equity. The main multi-asset comparison is deliberately narrower. The equal-weight four-sleeve rule at 175% gross exposure has almost exactly the ACO annual volatility (25.31% versus 25.28%). Its

Table 3: Corrected baseline results

Portfolio	Annual volatility	Equivalent saving	Retirement ruin
Domestic equity	23.34%	18.60%	15.52%
ACO 33/67	25.28%	10.00%	6.98%
Stocks/I 50/50	22.59%	10.56%	7.78%
Proportional four-sleeve, 175%	26.47%	6.26%	2.25%
Equal-weight four-sleeve, 175%	25.31%	7.16%	2.79%

Note: 10,000 paired lifecycle simulations. Equivalent saving is the contribution rate that matches expected utility of ACO 33/67 at a 10% saving rate. The four sleeves are ACO equity, fixed-notional currency-hedged global bonds, managed futures, and gold. The proportional rule is not an exact volatility match and is reported as a nearby, fixed-rule comparison.

simulated ruin is 2.79% rather than 6.98%, and its utility-equivalent saving rate is 7.16% rather than 10%. The paired 95% Monte Carlo interval for the ruin difference is $[-4.67, -3.71]$ percentage points.

5.2 Capital preservation without portfolio borrowing

At 100% gross exposure, both diversified families improve capital preservation without a portfolio-level loan. The equal-weight rule has 14.96% annual volatility and 6.21% ruin, against 25.28% and 6.98% for ACO. The proportional rule has 15.59% volatility and 5.22% ruin. These volatilities are about 60% of the benchmark’s full-panel value. With $G = 1$, the financing-spread term $\max(G - 1, 0)\phi$ is zero, so these results do not depend on that spread. Fees, hedge costs and the derivative exposures inside managed futures remain.

The utility comparison differs. Equivalent saving rises to 14.00% for the equal-weight rule and 12.82% for the proportional rule, against 10% for ACO. Lower ruin therefore does not imply higher expected utility: the latter also values retirement consumption and bequests. In these fixed families, capital preservation improves before borrowing; additional exposure is needed to reach benchmark utility at the same saving rate. This describes the tested rules, not a general necessity for leverage.

5.3 Historical windows and concentration of variance

We retain 1927–2025 as the primary sample. Large currency movements in the JST observations for Italy 1942 and France 1946 propagate into converted foreign returns. Their arithmetic origin is identifiable, but neither an error nor historical executability has been established. Extreme returns alone do not justify deletion.

Italy 1942 accounts for 48.35% of ACO’s sum of squared deviations from its full-panel mean, and 53.26% for the equal-weight 175% rule. Deleting this common row solely as a diagnostic reduces their volatilities to 18.17% and 17.30%, from 25.28% and 25.31%. Their respective top-five concentration shares are 54.61% and 65.38%; those sets need not contain identical rows. Thus the near equality of full-panel volatility is fragile and does not establish equal lifecycle risk.

Table 4 applies the three influence diagnostics consistently. A source event is removed from the resident bootstrap row and from every reconstructed international-equity and global-bond basket; the country diagnostic removes Italy from those baskets in all years and from the resident rows. The ranking remains: under the Italy-1942 event removal, the equal-weight 175% rule has 2.61% ruin and a 7.02% equivalent saving rate, versus 7.19% and 10% for ACO. Its annual volatility is 17.30%, against 18.17% for ACO. Removing France 1946 as well, or Italy in every year, changes the levels but not the direction of the comparison. These exercises show influence, rather than evidence that the excluded observations are errors or that their markets were unavailable to investors.

Table 4: Source-event and source-country influence diagnostics

Diagnostic	Portfolio	Volatility	Ruin	Equivalent saving
Italy 1942 event	ACO	18.17%	7.19%	10.00%
	Equal-weight 175%	17.30%	2.61%	7.02%
Italy 1942; France 1946 events	ACO	17.50%	7.17%	10.00%
	Equal-weight 175%	15.61%	2.74%	7.03%
Italy, all years	ACO	18.08%	7.54%	10.00%
	Equal-weight 175%	17.31%	2.70%	7.16%

10,000 paired simulations per diagnostic. The corresponding 95% paired Monte Carlo intervals for the equal-weight ruin difference are $[-5.06, -4.10]$, $[-4.91, -3.95]$, and $[-5.33, -4.35]$ percentage points. Full definitions, commands and input/code hashes are in `results/method_review/source_exclusions/provenance.json`.

The two later windows in Table 5 remove entire early regimes. They were selected during the audit and are sensitivities, not independent validation samples. The fixed 200% rules retain lower ruin and equivalent saving in each window, while their volatility relative to ACO changes. A common seed pairs portfolios within a run, not lifetime draws across different historical samples.

Table 5: Fixed allocations across historical windows

Window	Portfolio	Volatility	Ruin	Equivalent saving
1927–2025	ACO	25.28%	6.98%	10.00%
1927–2025	Proportional 200%	30.19%	1.99%	5.09%
1927–2025	Equal-weight 200%	28.86%	2.60%	5.92%
1950–2025	ACO	16.59%	5.42%	10.00%
1950–2025	Proportional 200%	16.67%	0.43%	5.53%
1950–2025	Equal-weight 200%	15.92%	0.72%	6.93%
1970–2025	ACO	17.24%	6.67%	10.00%
1970–2025	Proportional 200%	16.78%	0.16%	3.71%
1970–2025	Equal-weight 200%	16.16%	0.21%	3.95%

10,000 paired simulations per window; 1,561, 1,213 and 894 country-years, respectively. Fixed-notional hedges and baseline costs throughout.

5.4 Exposure and implementation sensitivity

The 200% endpoints improve the simulated outcomes further, but their risk is not comparable to the benchmark on the corrected raw panel: the proportional and equal-weight rules have annual

Table 6: Implementation checks

Specification	ACO ruin	Proportional 200% ruin	Equal-weight 200% ruin
Corrected baseline	6.98%	1.99%	2.60%
Legacy bootstrap end treatment	7.38%	1.98%	2.60%
Ideal full-value hedge	6.98%	1.98%	2.60%
Legacy return screen (diagnostic only)	7.38%	1.68%	2.32%
Leave-one-source-country-out range	7.10–8.66%	1.06–3.18%	1.72–3.54%

Note: First four rows use 10,000 paired simulations; the leave-one-out range uses 5,000 per omission. The 200% columns assess robustness of the portfolio ordering; their volatility exceeds that of ACO in the corrected baseline.

volatility of 30.19% and 28.86%, respectively. They are therefore sensitivity results, not evidence for an equal-risk claim. At that exposure, their ruin rates are 1.99% and 2.60%, with equivalent saving rates of 5.09% and 5.92%.

Table 6 records the checks most relevant to the construction. Reinstating the former return screen makes volatility look much lower, which explains the earlier apparent 200% match; because the screen depends on realized returns, it is excluded from the baseline. Reverting to the former bootstrap end treatment or to the idealized hedge changes levels modestly but does not reverse the ranking. A true leave-one-source-country-out exercise removes the country from resident observations and from the global equity and bond baskets. Across the 16 omissions, benchmark ruin ranges from 7.10% to 8.66%, while the two 200% sensitivity endpoints remain below it (ranges 1.06–3.18% and 1.72–3.54%). This supports robustness of the ordering, while also showing that its absolute risk depends on the historical sample.

At a 100-basis-point financing spread, the 200% proportional and equal-weight sensitivity rules have ruin of 2.65% and 3.53%, respectively; at 200 basis points this rises to 4.28% and 5.70%. These are mechanical cost stresses applied to the historical simulation. They are useful bounds, but do not establish the terms or capacity available to an investor.

6 Reproducibility and scope

Full-panel numerical outputs are stored in `results/main_core_n10000.json` and `results/main_ladders_n10000.json`. The historical-window outputs, commands and SHA-256 hashes of inputs and code are recorded under `results/method_review/sample_windows/`. A repeat of the post-1950 run reproduced its complete JSON exactly. The script `build/variance_concentration.py` checks baseline volatility agreement before reporting concentration and deletion diagnostics.

Reproduction from the supplied monthly managed-futures snapshot differs from reconstruction of that snapshot from market histories. The latter currently requires upstream code and inputs outside this standalone folder. Proprietary external benchmark series are not redistributed. The package therefore does not yet provide a complete independent reconstruction of every source series. Historical proxy quality, currency convertibility and intrayear funding remain substantive uncertainties.

7 Conclusion

In this public-data annual lifecycle model, specified portfolios combining international equities, global bonds, gold and managed futures have lower simulated ruin and utility-equivalent saving than the equity benchmark at the reported exposure levels. The ranking persists in the full 1927–2025 panel and the two later historical windows. This supports examining a broader opportunity set when comparing fixed lifecycle allocations.

The annual-volatility comparison is less stable. Extreme currency conversions dominate full-panel variance, so its near equality at one exposure level should not be interpreted as equal overall risk. We retain those observations without asserting that their historical implementation was feasible, and show their influence explicitly. The later windows are sensitivities selected after the audit, not independent confirmation.

Annual timing, reconstructed returns, assumed financing terms, and the absence of an instrument-level implementation limit extrapolation. The findings are conditional portfolio comparisons, not a universally optimal allocation or a forecast of retirement failure probabilities.

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A Asset-Class Return Histories

The cumulative-return exhibit reports annual U.S.-resident returns under the current fixed-notional hedge. The logarithmic axis makes proportional changes comparable across dates. Historical prices do not establish that every modeled exposure was available at the assumed cost.

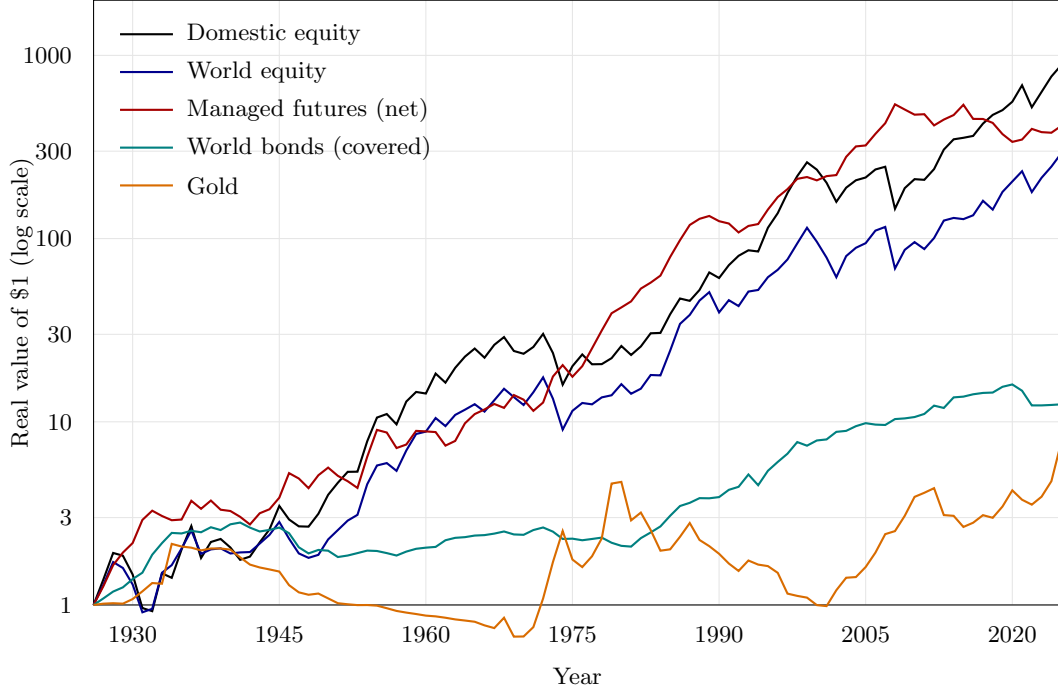


Figure 1: Cumulative real return by asset class, U.S. resident, 1927–2025. Managed futures are the reconstructed time-series-momentum proxy net of the 0.85% fee and 0.57% turnover cost. World bonds are the covered equal-weight basket of all panel sovereigns. Gold is the spliced MeasuringWorth–LBMA price less custody. Equity and gold are unhedged, while bonds and managed futures are currency-hedged under covered interest parity. The managed-futures proxy compounds at close to the equity rate over the century, but its decade-by-decade path is far more volatile before 1960, when the underlying market set is thin.

B Properties of the reconstructed managed-futures proxy

Managed futures enter through a reconstructed rule, so the construction and its external comparisons deserve separate scrutiny. The rule combines momentum signals at several horizons and scales positions using lagged volatility. The economic motivation follows Moskowitz, Ooi, and Pedersen (2012) and Hurst, Ooi, and Pedersen (2017). Agreement with a commercial index is a descriptive check; it cannot establish an investable historical track record.

B.1 Monthly inputs and the annual simulation series

The annual lifecycle panel and the monthly signal inputs are distinct data layers. The monthly snapshot supplies the managed-futures construction; complete calendar years are then compounded for the annual simulation. The source table records the original reconstruction’s inputs and trans-

formations. Price-only equities, synthetic bond returns, omitted commodity carry and roll, and incomplete early currency coverage limit the economic interpretation of the reconstructed return.

The annual simulation does not resolve intrayear trading, drawdowns, or margin requirements. Its revised currency treatment applies a forward to the initial foreign principal. Exchange-rate exposure on the subsequent asset gain or loss remains. This differs from removing foreign collateral and replacing it with the resident bill for the entire terminal asset value.

Table 7: Monthly data used to construct the managed-futures proxy

Input	Monthly source and transformation	What enters the MF rule
Equities	National price indexes from OECD Main Economic Indicators via FRED, and before their availability, the U.S., U.K., German, and French NBER Macrobistory monthly price series.	18 local-currency price returns. Dividends are absent.
Rates	Monthly sovereign yields from OECD via FRED, augmented before their start by NBER U.S. yields, U.K. Consol/Bank of England yields, and the French 3% rente. Each yield pair is priced as a par ten-year bond, then the lagged three-month cash return is subtracted.	18 synthetic local-currency bond excess returns, not changes in yields and not observed futures returns.
Commodities	NBER/BLS monthly wholesale prices for the early history and the World Bank Pink Sheet from 1960. Gold uses the official U.S. price before 1960 and the Pink Sheet thereafter, and silver starts in 1960.	19 commodity spot-price returns plus gold and silver. Carry, basis, and futures roll are unavailable and therefore set to zero.
Currencies	Board of Governors H.10 daily dollar quotations via FRED, reduced to the last available observation in each month, with a lagged three-month interest differential forming a synthetic forward excess return.	AUD, CAD, CHF, GBP, JPY from 1971 and EUR from 1999. Earlier European currencies are not backfilled.
Collateral and deflator	Monthly three-month cash rates from OECD via FRED, with a lagged prior-year JST/GMD fallback only where monthly rates do not exist, plus monthly U.S. CPI for deflation.	U.S. cash collateral is added once to the USD MF NAV, and CPI converts the final monthly USD return to real terms.

Note: The machine-readable source-segment file records the provider, series identifier, dates, transformation, and source seam for every market. The source list is intentionally more granular than the annual Macrobistory panel: the latter cannot identify a month-end trading signal. The monthly snapshot is frozen for this replication, and source seams receive no return in the first month of a new segment, so a signal cannot cross them silently.

B.2 Comparison with commercial trend-following indexes

These monthly correlations use the supplied dollar proxy and external return series. They do not depend on the resident panel, annual hedge or lifecycle bootstrap; those revisions do not invalidate

this descriptive evidence. The external comparison places the net monthly proxy alongside SG CTA, SG Trend, Barclay BTOP50, and the KMLMSIM simulation over their common period. The indexes are external benchmarks and enter no lifecycle resampling. Their proprietary inputs remain subject to their source terms. The monthly figure and correlations show co-movement, not replication of a fund or validation of the pre-index history.

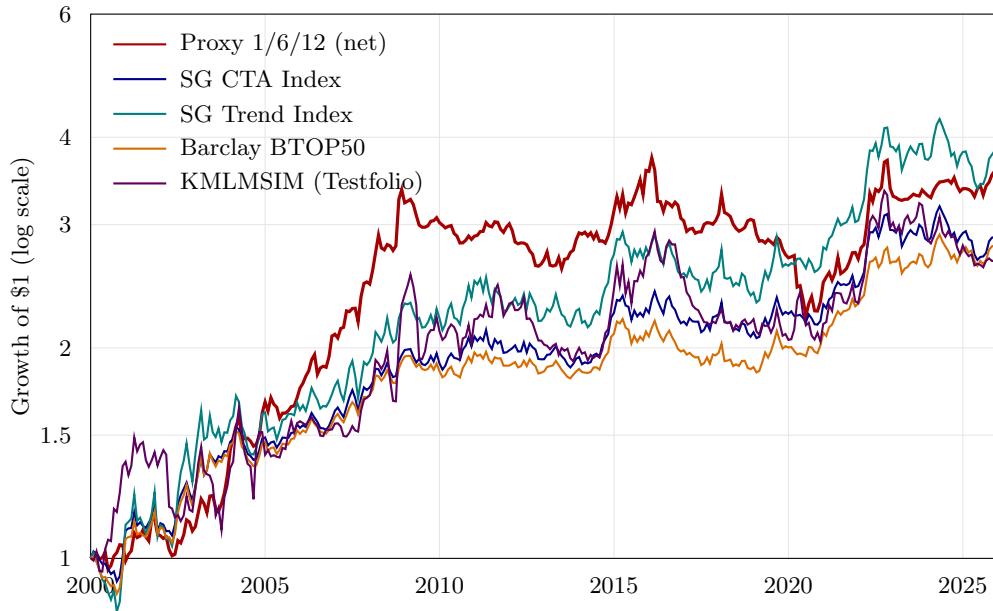


Figure 2: Growth of one dollar, monthly, 2000–2025: the net managed-futures proxy (variant 1/6/12, the one used in the paper) against three commercial trend-following indexes and KMLMSIM, the Testfolio simulation of KMLM. SG CTA, SG Trend, and BTOP50 are net of index fees, and the proxy is net of the 0.85% management fee and 0.57% turnover cost. KMLMSIM uses the KFA MLM Index less 0.90% per year before the ETF’s inception. The SG and BarclayHedge indexes are proprietary and are used here only under the terms held for this research.

Table 8: Monthly correlation of the net proxy with trend-following benchmarks

Benchmark	Corr.	Common window
SG CTA Index	0.51	2000-01–2025-12 (312 mo.)
SG Trend Index	0.49	2000-01–2025-12 (312 mo.)
Barclay BTOP50 Index	0.53	2000-01–2025-12 (312 mo.)
KMLMSIM (Testfolio simulation)	0.52	2000-01–2025-12 (312 mo.)
DBMF ETF (live)	0.61	2019-06–2025-12 (79 mo.)
KMLM ETF (live)	0.63	2021-01–2025-12 (60 mo.)

Memo, 2000–2025: the three commercial indexes correlate 0.96–0.97 with each other, whereas KMLMSIM correlates 0.60 with SG CTA. The proxy near 0.51 sits between these two reference patterns.

Note: contemporaneous Pearson correlation of monthly returns over the common months shown. KMLMSIM is Testfolio’s KFA MLM Index-based simulation of KMLM before the ETF’s inception, and the “live” rows use the ETFs’ own return history from the first full month after listing.

B.3 The proxy among commercial benchmarks and investable funds

The broader comparison retains the original pairwise correlation matrix, the matrix on a common window, and the regressions on the proxy. Pairwise windows differ when funds have different inception dates. The common-window matrix addresses that comparability issue, at the cost of a shorter history. These exhibits are descriptive external comparisons; a correlation, regression coefficient, or insignificant intercept does not make the proxy interchangeable with an investable product.

Table 9: Correlation matrix of monthly returns: the net proxy, commercial benchmarks, and investable managed-futures funds

	SG CTA	SG Trend	BTOP50	KMLMSIM	DBMF	KMLM	QMHIX	AHLIX	AHLT	IMF	ISMF	WTMF	APEX
Proxy	0.51	0.49	0.53	0.52	0.61	0.63	0.52	0.54	0.65	0.63	0.58	0.23	0.15
SG CTA	—	0.97	0.97	0.60	0.86	0.77	0.83	0.84	0.86	0.76	0.87	0.40	0.50
SG Trend	0.97	—	0.96	0.60	0.86	0.74	0.85	0.86	0.85	0.72	0.89	0.42	0.46
BTOP50	0.97	0.96	—	0.58	0.85	0.73	0.79	0.83	0.84	0.69	0.82	0.41	0.57
KMLMSIM	0.60	0.60	0.58	—	0.66	1.00	0.75	0.63	0.69	0.89	0.86	0.34	0.33
DBMF	0.86	0.86	0.85	0.66	—	0.68	0.73	0.74	0.77	0.29	0.53	0.24	0.50
KMLM	0.77	0.74	0.73	1.00	0.68	—	0.74	0.65	0.69	0.89	0.86	0.12	0.33
QMHIX	0.83	0.85	0.79	0.75	0.73	0.74	—	0.70	0.74	0.54	0.82	0.33	0.40
AHLIX	0.84	0.86	0.83	0.63	0.74	0.65	0.70	—	0.85	0.76	0.83	0.39	0.28
AHLT	0.86	0.85	0.84	0.69	0.77	0.69	0.74	0.85	—	0.82	0.90	0.13	0.56
IMF	0.76	0.72	0.69	0.89	0.29	0.89	0.54	0.76	0.82	—	0.86	0.28	0.40
ISMF	0.87	0.89	0.82	0.86	0.53	0.86	0.82	0.83	0.90	0.86	—	0.42	0.59
WTMF	0.40	0.42	0.41	0.34	0.24	0.12	0.33	0.39	0.13	0.28	0.42	—	0.01
APEX	0.50	0.46	0.57	0.33	0.50	0.33	0.40	0.28	0.56	0.40	0.59	0.01	—

Note: contemporaneous Pearson correlation of monthly returns, each pair over the overlap of its two histories within 2000-01 to 2025-12, the frame of the appendix. The first month in that frame is 2000-01 for the SG indexes (their full history), 2019-06 for DBMF, 2021-01 for KMLM, 2011-02 for WTMF, 2013-08 for QMHIX, 2014-09 for AHLIX, 2022-04 for Apex, 2023-10 for AHLT, and 2025-04 for IMF and ISMF; BTOP50 (from 1987) and KMLMSIM (from 1988) are restricted to the common frame, and the proxy is available from 1926-03, so the number of overlapping months of a pair is determined by the later of its two start months. The index returns are the proprietary SG and BarclayHedge series and the testfolio series of Table 8; fund returns are total returns net of fund fees from Yahoo Finance adjusted closes. Shading is proportional to the correlation; the horizontal rule separates the benchmark block from the investable funds. QMHIX, AHLIX, AHLT, IMF, ISMF, DBMF, and KMLM are pure trend-following products; WTMF is a systematic equity-timing strategy and Apex a multi-strategy fund.

Table 10: Correlation matrix over a common window: the proxy and the benchmarks and funds with at least ten years of history

	SG CTA	SG Trend	BTOP50	KMLMSIM	WTMF	QMHIX	AHLIX
Proxy	0.57	0.57	0.60	0.56	0.23	0.52	0.54
SG CTA	—	0.98	0.97	0.72	0.39	0.84	0.84
SG Trend	0.98	—	0.96	0.72	0.40	0.86	0.86
BTOP50	0.97	0.96	—	0.69	0.41	0.79	0.83
KMLMSIM	0.72	0.72	0.69	—	0.31	0.75	0.63
WTMF	0.39	0.40	0.41	0.31	—	0.33	0.39
QMHIX	0.84	0.86	0.79	0.75	0.33	—	0.70
AHLIX	0.84	0.86	0.83	0.63	0.39	0.70	—

Note: contemporaneous Pearson correlation of monthly returns, every pair evaluated over the same window 2014-09 to 2025-12 (136 months), the maximal common window of the eight series (proxy from 1926-03; SG indexes from 2000-01, BTOP50 and KMLMSIM restricted to the common frame; WTMF from 2011-02, QMHIX from 2013-08, AHLIX from 2014-09). DBMF (79 months) and KMLM (60 months) do not meet the ten-year rule. Sources and conventions as in Table 9.

Table 11: Regressions of managed-futures benchmarks and investable funds on the net proxy

Series	n	α (%/mo.)	β	R^2	σ_f/σ_p	TE (%/yr.)
SG CTA	312	+0.18 (+1.4)	0.41 (+10.6)	0.26	0.80	9.8
SG Trend	312	+0.22 (+1.2)	0.60 (+9.9)	0.24	1.24	12.4
BTOP50	312	+0.18 (+1.6)	0.39 (+10.9)	0.28	0.74	9.5
KMLMSIM	312	+0.06 (+0.3)	0.76 (+10.9)	0.28	1.44	13.5
DBMF	79	+0.45 (+1.6)	0.62 (+6.8)	0.38	1.00	9.8
KMLM	60	−0.13 (−0.3)	0.78 (+6.1)	0.39	1.24	10.2
QMHIX	149	+0.30 (+0.9)	0.73 (+7.3)	0.27	1.42	13.4
AHLIX	136	+0.28 (+1.4)	0.47 (+7.4)	0.29	0.87	10.1
AHLT	27	−0.53 (−0.8)	1.49 (+4.3)	0.42	2.30	12.0
IMF	9	−2.08 (−1.5)	1.10 (+2.2)	0.40	1.73	11.9
ISMF	9	+0.57 (+0.9)	0.43 (+1.9)	0.34	0.74	7.3
WTMF	179	+0.04 (+0.3)	0.14 (+3.1)	0.05	0.62	10.7
APEX	45	+1.11 (+2.9)	0.13 (+1.0)	0.02	0.87	12.5

Note: ordinary least squares $R_t^s = \alpha + \beta R_t^{proxy} + \varepsilon_t$ on monthly returns, each series over its overlap with the proxy within 2000-01 to 2025-12 (n column; all windows end in 2025-12). α is in per cent per month; α and β carry Student t statistics in parentheses. σ_f/σ_p is the ratio of annualised volatilities, and TE the tracking error: the annualised standard deviation of the active return $R_t^s - R_t^{proxy}$, in per cent per year. Sources and conventions as in Table 9.

B.4 Annual properties against the other asset classes

The annual moments and scatter plots use the current U.S.-resident panel and fixed-notional hedge. The proxy is net of a 0.85% management fee and a 0.57% annual turnover deduction. These descriptive fits measure association; they do not identify a causal mechanism for the reduction in retirement ruin.

Table 12: The managed-futures proxy against the other asset classes

Asset	Mean	SD	Sharpe ₀	Skew	5th pctl	Max DD
Domestic equity	8.81%	18.69%	0.47	-0.24	-21.4%	-51.9%
World equity	7.42%	17.48%	0.42	-0.25	-23.1%	-48.1%
World bonds (covered)	2.82%	7.03%	0.40	+0.15	-7.6%	-35.2%
Gold	3.70%	19.70%	0.19	+1.88	-21.2%	-79.0%
Managed-futures proxy (net)	7.17%	14.05%	0.51	+0.44	-13.1%	-37.6%

Year	World equity	MF proxy	Gold	U.S. inflation
1930	-18.7%	12.7%	6.0%	-2.3%
1931	-29.5%	33.8%	10.2%	-9.0%
1937	-25.4%	-9.6%	-3.4%	3.6%
1946	-18.9%	36.3%	-15.7%	8.3%
1973	-23.5%	38.6%	58.1%	6.3%
1974	-32.2%	15.4%	47.6%	11.0%
1990	-22.5%	-6.3%	-9.2%	4.9%
2002	-21.8%	1.4%	22.0%	1.6%
2008	-40.7%	26.7%	3.9%	3.8%
2022	-23.1%	14.4%	-6.0%	8.0%

Note: U.S. resident, annual real returns, 1927–2025. *Top panel:* Sharpe _{$r=0$} is the mean over the standard deviation with a zero real cash rate, and “Max DD” is the largest peak-to-trough decline in cumulative real value. *Bottom panel:* the ten worst years for world equity and the proxy’s realized return in each, where “U.S. inflation” is the same year’s CPI change.

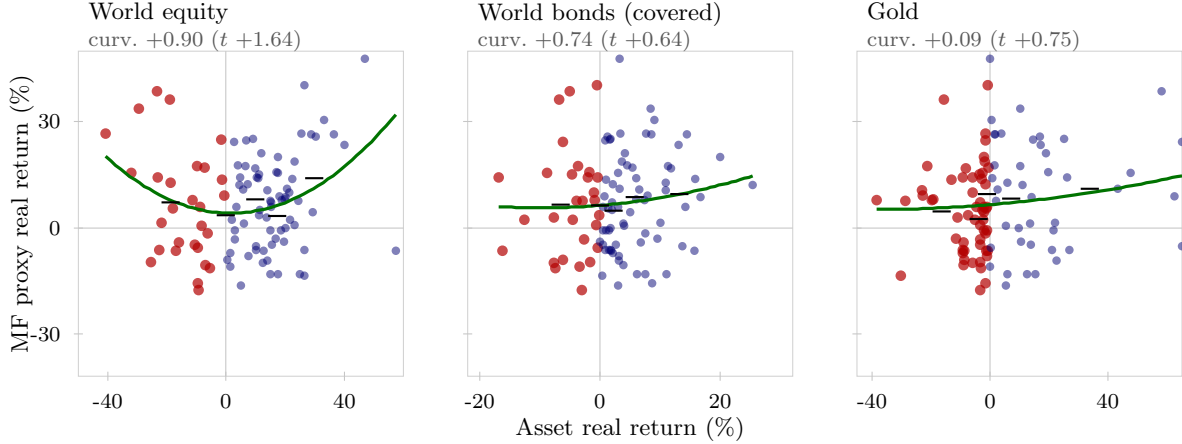


Figure 3: Annual real return of the net managed-futures proxy against the annual real return of each asset class, U.S. resident, 1927–2025. Each point is one year, with red points the years the asset itself lost and blue points the years it gained. The green curve is a descriptive quadratic fit and the black ticks are the mean proxy return within each quintile of the asset’s return. “curv.” is the coefficient on the squared term, and t is its heteroskedasticity-robust (White HC1) statistic.

B.5 Sensitivity to the construction choices

The series-level comparison varies signal horizons and management fees, using each variant’s measured turnover cost. It describes the underlying proxy series independently of the lifecycle bootstrap

and resident-currency hedge.

Table 13: Managed-futures proxy under alternative construction choices

Variant	Mean	SD	Sharpe ₀	Skew	5th pctl	Corr. baseline
1/6/12 blend (baseline) [†]	7.04%	13.93%	0.51	+0.39	-12.1%	1.000
1/3/12 blend	7.25%	14.33%	0.51	+0.60	-13.1%	0.961
12-month signal only	6.96%	13.76%	0.51	+0.04	-17.4%	0.831
1-month signal only	5.21%	12.74%	0.41	+0.66	-12.5%	0.772
1/6/12 blend, fee doubled to 1.70%	6.13%	13.81%	0.44	+0.39	-12.9%	1.000

Note: U.S. resident, annual real returns, 1927–2025, net of each variant’s management fee and its own realized turnover cost. “Corr. baseline” is the Pearson correlation of the annual series with the 1/6/12 blend used throughout the paper. The baseline mean here differs from Table 12 by about 0.15 point because that table applies a flat annual turnover drag while this one compounds the monthly realized cost. The correlations are unaffected by the cost convention. [†]Baseline.

C Leverage ladders

The following tables use the revised 1,561-country-year panel, the ACO-style block continuation, and the fixed-notional forward convention. Each family has five specified gross-exposure levels. Volatility is measured after the weights are set; the tables do not estimate a volatility target. They use 10,000 paths at seed 20260827 and the same paired ACO benchmark. The benchmark has 25.28% annual volatility, 6.98% retirement ruin, and a 10% reference saving rate. Because a small number of observations dominate the pooled variance, similar volatility is an especially fragile description of risk in this sample.

Table 14: Proportional leverage ladder: revised results

Gross	Equity/Bonds/Gold/MF	Volatility	Saving	Ruin
100%	40/26.67/16.67/16.67	15.59%	12.82%	5.22%
125%	50/33.33/20.83/20.83	19.15%	9.90%	3.39%
150%	60/40/25/25	22.78%	7.81%	2.66%
175%	70/46.67/29.17/29.17	26.47%	6.26%	2.25%
200%	80/53.33/33.33/33.33	30.19%	5.09%	1.99%

Note: Equity is the ACO 33/67 sleeve. Bonds and managed futures use the revised fixed-notional hedge. Saving is the contribution rate that matches benchmark utility. Source: `results/main_ladders_n10000.json`.

Table 15: Equal-weight leverage ladder: revised results

Gross	Equity/Bonds/Gold/MF	Volatility	Saving	Ruin
100%	25/25/25/25	14.96%	14.00%	6.21%
125%	31.25/31.25/31.25/31.25	18.33%	10.95%	4.20%
150%	37.5/37.5/37.5/37.5	21.80%	8.78%	3.27%
175%	43.75/43.75/43.75/43.75	25.31%	7.16%	2.79%
200%	50/50/50/50	28.86%	5.92%	2.60%

Note: Equity is the ACO 33/67 sleeve. Bonds and managed futures use the revised fixed-notional hedge. Saving is the contribution rate that matches benchmark utility. Source: `results/main_ladders_n10000.json`.

D Historical-sample uncertainty

Monte Carlo intervals measure sampling noise conditional on the observed history and the specified model. They do not measure uncertainty about the historical return-generating process. We retain 1927–2025 as the main sample. The revised 1950–2025 and 1970–2025 experiments, each with 10,000 paths, address sensitivity to the historical window. They are reported in the main text and stored under `results/method_review/sample_windows/`. These overlapping periods are selected historical comparisons, not independent validation samples.

The revised source-country omissions remove the country from resident rows, international-equity constituents, and world-bond constituents. At 5,000 paths per omission, ACO ruin ranges from 7.10% to 8.66%; the proportional 200% family ranges from 1.06% to 3.18%, and the equal-weight 200% family from 1.72% to 3.54%. The corresponding saving rates range from 5.00% to 5.26% and from 5.63% to 6.16%, against each sample’s 10% reference. These are ranges across omissions, not confidence intervals. The source audit is `paper/figures/source_country_sensitivity.json`.

The separate event experiments remove Italy 1942, then Italy 1942 and France 1946; the country experiment removes Italy throughout. Their reconstructed inputs and outputs are under `results/method_review/source_exclusions/`. They measure the influence of observations identified during the audit. They do not establish that those observations are errors, and their exclusions do not reconstruct every underlying monthly managed-futures input. The main text reports these results.

E Risk-aversion sensitivity

The preference exercise holds the two 200% portfolio rules fixed. Risk aversion takes six values: 2, 3, 3.84, 5, 7.5, and 10. The bequest calibration changes with γ as recorded in the table. This is a joint preference sensitivity under that calibration, not an optimized allocation at each risk aversion. Retirement ruin is a depletion event and therefore does not change when preferences alone change. Equivalent saving does change. The revised results keep both families below the benchmark’s saving rate throughout the reported grid. The 200% portfolios do not have the benchmark’s full-panel volatility.

Table 16: Risk-aversion sensitivity of the fixed diversified families

γ	ACO ruin	Proportional 200%		Equal-weight 200%	
		Ruin	Equiv. saving	Ruin	Equiv. saving
<i>lower is better, ACO equivalent saving is 10% at every γ</i>					
2	6.98%	1.99%	2.38%	2.60%	3.23%
3	6.98%	1.99%	3.73%	2.60%	4.59%
3.84 [†]	6.98%	1.99%	5.09%	2.60%	5.92%
5	6.98%	1.99%	4.90%	2.60%	5.73%
7.5	6.98%	1.99%	4.64%	2.60%	5.48%
10	6.98%	1.99%	4.53%	2.60%	5.38%

Note: 10,000 paired lifecycle paths per row, full 1,561-country-year panel, ten-year mean stationary blocks, $\phi = 30$ bp and $\kappa = 10$ bp. Entries are levels, in percent: retirement ruin and the utility-equivalent saving rate, both lower is better. The ACO 33/67 ruin column is shown for reference and its equivalent saving is 10% throughout. The two 200% families keep their fixed weights and gross exposure at every γ , so this is a sensitivity of the fixed rule, not a reoptimized allocation. Retirement ruin is a wealth-depletion event and does not depend on γ , so the ruin columns are constant by construction and only the equivalent-saving columns respond. The bequest intensity is rescaled to each γ as $\theta(\gamma) = \theta_{3.84} \cdot 12^{3.84-\gamma}$, matching ACO's monthly-to-annual convention.

[†]Baseline.

F Contribution- and withdrawal-rate sensitivity

The revised policy exercise varies contribution rates at a fixed 4% withdrawal rule and withdrawal rates at a fixed 10% benchmark contribution. On the contribution axis, the comparison is anchored to benchmark utility at that row's contribution rate. On the withdrawal axis, it is anchored to the benchmark under that row's withdrawal rule. Thus saving rates should be compared within a row.

With proportional contributions and an initial-wealth-based withdrawal rule, changing the contribution rate scales account wealth and withdrawals together. The unchanged ruin rates on that axis are a property of the model, not evidence that saving has no economic importance. Higher withdrawal rates increase depletion in all three portfolios. Both diversified families retain lower simulated ruin and equivalent saving over this grid; the exercise remains conditional on the fixed 200% allocations and the annual return model.

Table 17: Contribution- and withdrawal-rate sensitivity of the fixed diversified families

Rate	ACO 33/67		Proportional 200%		Equal-weight 200%	
	Ruin	Saving	Ruin	Equiv. saving	Ruin	Equiv. saving
<i>lower is better, the ACO columns are the reference</i>						
<i>Contribution rate r_c (withdrawal held at 4%)</i>						
$r_c = 5\%$	6.98%	5.00%	1.99%	2.67%	2.60%	3.07%
$r_c = 10\%^\dagger$	6.98%	10.00%	1.99%	5.09%	2.60%	5.92%
$r_c = 15\%$	6.98%	15.00%	1.99%	7.36%	2.60%	8.62%
<i>Withdrawal rate r_w (contribution held at 10%)</i>						
$r_w = 3\%$	2.54%	10.00%	0.95%	5.48%	1.16%	6.32%
$r_w = 4\%^\dagger$	6.98%	10.00%	1.99%	5.09%	2.60%	5.92%
$r_w = 5\%$	15.47%	10.00%	4.81%	4.34%	6.40%	5.12%

Note: 10,000 paired lifecycle paths per row, full 1,561-country-year panel, ten-year mean stationary blocks, $\phi = 30$ bp and $\kappa = 10$ bp. Entries are levels, in percent: retirement ruin and the utility-equivalent saving rate, both lower is better. The two ACO 33/67 columns are the reference. The two 200% families keep their fixed weights and gross exposure in every row. On the contribution-rate axis the ACO 33/67 benchmark also contributes r_c , so its reference saving rate is r_c itself and the family saving is the rate that matches ACO's lifecycle utility at that r_c . On the withdrawal-rate axis the benchmark saving rate stays at 10%. † Baseline.

G Reproducibility and Conversion Checks

The revised baseline is identified by 1,561 resident country-years, 10,000 paths, seed 20260827, ten-year mean blocks, and the `fixed_notional` hedge convention. Source files, configurations, and outputs must be checked together: sharing a seed does not make paths identical across different historical panels.

The annual conversion and simulation can be reproduced from the supplied annual inputs. The monthly managed-futures reconstruction has an upstream engine dependency outside this paper directory. Accordingly, the package's ability to consume the frozen monthly output must be distinguished from an autonomous reconstruction of that output from the original providers. The external-index comparisons also depend on their licensed inputs. Neither limitation is removed by a successful LaTeX compilation.

For a foreign asset's local real return R_f^L and inflation π_f , the nominal gross return is $(1 + R_f^L)(1 + \pi_f)$. A fixed-notional forward covers the initial foreign principal. If q is the realized resident-currency value of one unit of foreign currency relative to its initial value, G_f the foreign nominal gross asset return, and B_c/B_f the covered-interest-parity forward factor, the hedged nominal gross payoff is

$$G_{f \rightarrow c}^H = \frac{B_c}{B_f} + (G_f - 1)q.$$

Deflating by resident inflation produces the real return used in the revised panel. The second term retains currency exposure on the foreign gain or loss. Consequently, equality of all residents' asset excess returns is not a valid check for this convention. That identity belongs to the earlier idealized full-terminal-value hedge.

The return-cap diagnostic, ideal hedge, legacy block restart, restricted historical windows, and

source exclusions are distinct experiments. Their output files identify the changed convention. The revised model does not silently discard a return because it is large. Documented market-year exclusions and influence diagnostics have different interpretations and should remain separately identifiable.

H Margin calls on the do-it-yourself route

The earlier margin exercise modeled annual rebalancing and a 25% maintenance threshold on a directly financed account. Its reported absence of year-end calls was a result of that earlier annual sample and convention. It is not a revised estimate of margin-call risk. The exercise cannot observe a breach and liquidation within the year.

For initial equity normalized to one, gross assets L , debt $L - 1$, asset-book nominal gross return G_A , and debt gross accrual G_D , the year-end equity fraction is

$$m_{\text{end}} = \frac{LG_A - (L - 1)G_D}{LG_A}.$$

A maintenance call at threshold m occurs when $G_A < (L - 1)G_D/[L(1 - m)]$, provided the asset value remains positive. The threshold therefore depends on financing as well as exposure. Nominal rather than real account values determine the margin test.

An annual check gives no bound on intrayear calls, variation margin, liquidity needs, changing broker requirements, or losses caused by forced deleveraging. Holding an unlevered position in a fund may move margin management inside the fund, but it does not remove the fund's financing, liquidity, or deleveraging risk. The lifecycle results are consequently conditional portfolio-return experiments, not an operational validation of a particular fund or brokerage implementation.